

Product Requirements & Data Specification

AI-Based Automatic Block Planning · Chennai Division

Purpose What the system must do, for whom, with what data, and how “done” is judged

Companion documents 1. System Blueprint · 3. Complete Project Compendium

Version 2.0 — requirements and data specification consolidated

How to use this document. Sections 1–6 are the product requirements: personas, user stories, functional and non-functional requirements, each with acceptance criteria. Sections 7–12 are the data specification — the build contract for the data layer. Every table in Section 8 onwards should exist as a CSV or database table before modelling or optimisation work begins.

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1 Vision, problem and success criteria

1.1 Problem statement

Engineering, Traction Distribution and Signal & Telecommunication each maintain fixed line-side assets. Each independently requests track access through BDMS, based on defects and overdue tasks held in TMS, SMMS and TDMS. Nobody merges these requests. A section that could have been taken once for three hours is taken three separate times, each outage costing train delays, while genuinely urgent defects wait because the corridor is already spoken for. Meanwhile emergencies consume windows unpredictably.

1.2 Why now

The practice was harmless under slack. Train density has risen, freight has grown, timetables have tightened and free corridor windows have shrunk. **Block availability has become the scarce resource.** The scarcity changed; the planning process did not.

1.3 Vision

Pool all three departments' pending work into one scored, costed view; match it against real corridor availability; pack compatible co-located jobs into shared blocks; route the result through the people who already hold authority; and re-plan when reality breaks the schedule.

1.4 Success criteria

Criterion	Target	Measured how
Line occupation reduction	Material reduction versus baseline	Counterfactual comparison, 30 seeded weeks
Merge rate	Substantially above the incidental baseline	Blocks containing more than one department
Weighted detention reduction	Material reduction versus baseline	Objective function first term
Tasks completed	Not worse than baseline	Guards against scheduling nothing
Safety deadline breaches	Zero, always	Hard constraint violation count
Overrun rate	Consistent with the chosen buffer quantile	Actual versus sanctioned duration

2 Personas

Persona	Role	Needs from the system
P1 — Field staff SSE/JE, all three departments	Raises defects, executes blocks on site	To know what they are doing tonight, confirm readiness, start and stop work, and record what actually happened — at 02:30, one-handed, on weak signal
P2 — Departmental officer Sr.DEN, Sr.DEE/TrD, Sr.DSTE	Owns a department's backlog and approves its work	A ranked backlog with visible reasoning, and the ability to approve or reject proposals with a structured reason
P3 — Operations Sr.DOM, chief controller	Holds final sanction; declares emergencies	Traffic impact of every proposal, the ability to refuse, and instant re-planning when disruption occurs
P4 — All departments, jointly	Coordination across departmental boundaries	One shared view of who wants which corridor when, with conflicts explained by rule
P5 — Divisional management DRM	Oversight, never individual tasks	Trend indicators that imply an action, aggregated at team and section level only

3 User stories with acceptance criteria

3.1 Field staff (P1)

ID	Story	Acceptance criteria
US-01	As an SSE I raise a defect from the field so it enters the planning pool	Location captured to section, km and line; type and severity selected; photograph attached; record queues locally and syncs when signal returns
US-02	As an SSE I see only my own upcoming work	Single card view; no other department's tasks visible; readable at arm's length in poor light
US-03	As an SSE I confirm readiness 24 hours before a block	Crew, materials, machine and permissions each toggled; incomplete readiness raises an availment flag visible to the officer

US-04	As an SSE I record actual start and end times	One large control to start and stop; running timer against sanctioned end; overrun prompts a structured reason from tappable options; never blocked by absence of network
US-05	As an SSE I work in Tamil or English	Language switchable; all operational labels translated

3.2 Departmental officer (P2)

ID	Story	Acceptance criteria
US-06	As Sr.DEN I see my department's backlog ranked by risk	Sorted by composite risk score; days pending and location shown; refreshes on each planning run
US-07	As Sr.DEN I understand why a task ranked where it did	Score decomposes into at most five named factors with signed contributions, in plain language — not a statistical plot
US-08	As Sr.DEN I approve or reject proposals involving my staff	Rejection requires a reason from a fixed list; free text optional in addition; the system immediately shows the next-best alternative and its cost
US-09	As Sr.DEN I am not asked to re-approve unchanged scope	If another department withdraws and my department's scope is unaltered, my approval persists
US-10	As Sr.DEN I am warned when a safety deadline is at risk	Any safety-critical task the optimiser could not fit is badged prominently and escalated, never silently deferred

3.3 Operations (P3)

ID	Story	Acceptance criteria
US-11	As Sr.DOM I see every proposal against the timetable	Time-by-section grid; train density shown as background intensity; proposed blocks, corridor blocks and newly requested access visually distinct
US-12	As Sr.DOM I see the traffic cost of each proposal	Trains held, rerouted and cancelled, with predicted weighted detention minutes; rejected alternatives listed with their costs
US-13	As Sr.DOM I hold final refusal	Refusal available at any point before issue, with a recorded reason; no block reaches BDMS without operations sanction
US-14	As Sr.DOM I declare an emergency and receive a revised plan	Event logged with type, location, duration estimate and resources diverted; re-plan runs automatically; result presented as a diff , not a regenerated schedule
US-15	As Sr.DOM I trust that committed work is not silently moved	Blocks inside the freeze horizon are never re-optimised; those rendered infeasible are cancelled and surfaced with a reason

3.4 Coordination and management (P4, P5)

ID	Story	Acceptance criteria
US-16	As any officer I see all three departments' work on one corridor timeline	Department colour consistent across every screen; merged blocks show a lane per department
US-17	As any officer I see which task is setting the block duration	Critical path highlighted within the merged block
US-18	As any officer I understand why two tasks cannot share a block	Conflict shown inline, naming the compatibility rule identifier and its basis

US-19	As a DRM I see whether coordination is working	Headline figure of blocks merged and hours saved; merge rate trend; backlog trend; safety breaches called out separately
US-20	As a DRM I see the maintenance trade-off quantified	Detention caused by blocks plotted against detention caused by speed restrictions from undone maintenance, on one axis
US-21	As a DRM I learn about constraints nobody encoded	Proposals rejected three or more times flagged by name for review

4 Functional requirements

ID	Requirement	Detail
Integration		
FR-01	Ingest from three departmental systems	Scheduled pull from TMS, SMMS and TDMS, mapped onto the unified task schema
FR-02	Ingest corridor availability	Train paths, goods forecast and pre-planned corridor blocks from COA
FR-03	Read and write BDMS	Read existing requests to avoid double-booking; write sanctioned blocks as formal requests. BDMS remains the system of record; on conflict BDMS wins
FR-04	Deduplicate	Re-inspections of the same defect resolve to one task
Scoring		
FR-05	Predict task duration with uncertainty	Quantile outputs; buffer derived from the spread
FR-06	Estimate escalation hazard	Marginal daily hazard, calibrated, composable to any horizon
FR-07	Estimate train detention	Analytical estimate with learned residual correction
FR-08	Estimate block availment risk	Calibrated probability the sanctioned block is wasted
FR-09	Explain every score	Named factor contributions attached to each task
FR-10	Operate without training data	Three-stage fallback: rule-based, hierarchical, full model
Planning		
FR-11	Enumerate candidate windows	Corridor blocks plus detected traffic gaps, net of setup and clearance margins
FR-12	Generate merge candidates	Pairwise-compatible subsets up to configured size; singletons always included
FR-13	Compute merged duration by critical path	Honouring parallel and sequential relations, not sum or naive maximum
FR-14	Escalate access type on merge	Union of access types, with power-implies-traffic on electrified sections
FR-15	Produce a weekly plan	Optimised assignment plus local refinement within traffic gaps
FR-16	Produce a monthly plan	Corridor quotas, machine positioning, reserved capacity. Never pre-assigns tasks
FR-17	Flag newly requested access	Windows outside pre-planned corridor blocks marked distinctly and penalised in the objective
FR-18	Enforce safety deadlines as hard constraints	Never traded against cost; infeasibility escalates visibly
FR-19	Enforce crew rest and duty hours	Hard constraint, not a preference
FR-20	Enforce machine transit feasibility	Hard constraint; availment risk handles only the probabilistic case
FR-21	Never sever a section	Line-union feasibility enforced across parallel edges
Disruption		
FR-22	Accept manual emergency declaration	Type, location, duration estimate, resources diverted
FR-23	Re-plan automatically on declaration	Scoped to affected sections and neighbours; frozen blocks fixed
FR-24	Present changes as a diff	Moved, cancelled and unchanged counts, not a regenerated plan

FR-25	Generate emergency follow-up tasks	Permanent repair enters the pool with a short hard deadline
Workflow		
FR-26	Require approval from every department in a block	Merged blocks need all participating departments
FR-27	Degrade before re-solving on partial rejection	Drop the rejecting department, recompute; re-solve only if the reduced block no longer wins
FR-28	Capture structured rejection reasons	Fixed enumeration; free text optional in addition
FR-29	Apply proportionate approval policy	Explicit approval for safety-critical and new access; timed auto-approve for routine corridor work
FR-30	Audit every transition	Actor, timestamp, states, reason, plan version and optimiser cost figures at the moment of decision
FR-31	Flag persistent rejection	Three or more rejections surfaces the proposal to the DRM
Evaluation		
FR-32	Provide a baseline planner	Independent per-department planning with a configurable incidental merge rate
FR-33	Run seeded comparative experiments	Identical scenarios to both planners; mean and spread across 30 seeds
FR-34	Support component ablation	Each major component disableable by configuration flag

5 Non-functional requirements

ID	Area	Requirement
NFR-01	Weekly solve performance	Complete within the configured time limit for a full division; solve is queued, never synchronous to an HTTP request
NFR-02	Re-plan responsiveness	Emergency re-plan returns within a small number of seconds and pushes live to connected clients
NFR-03	Field app offline operation	All field actions queue locally and sync opportunistically; no action blocked by absence of network; sync state shown honestly
NFR-04	Explainability	Interpretable models preferred over higher-accuracy opaque ones; every officer-facing score decomposes
NFR-05	Role-based access control	Sanction, approve, execute and view are distinct permissions; enforced at the API, not the interface
NFR-06	No individual-level performance data	Execution data aggregates at team and section level only, enforced in the API. No leaderboards, rankings or crew comparisons anywhere
NFR-07	Auditability	Full history reconstructable, including what the officer saw at decision time
NFR-08	Configurability	Objective weights, horizons, buffers, thresholds and the corridor block pattern are configuration per division
NFR-09	Reproducibility	Fixed random seed across development, evaluation and demonstration
NFR-10	Graceful degradation	Missing models fall back to rules; unmergeable tasks fall back to singletons; a valid plan always results
NFR-11	Division agnosticism	<code>division_id</code> carried throughout so expansion is a retraining exercise, not a rewrite
NFR-12	Language	Tamil and English in the field application

6 Explicitly out of scope

Excluded	Reason
Rolling stock maintenance	Mechanical / C&W department, different systems, requires no block
Possessions and mega-blocks	Planned months ahead at zonal level; enter as fixed unavailable windows
Real-time train regulation	Control's on-the-day function, minute-by-minute; different system entirely
Interlocking and route conflict resolution	Requires route-table data absent from all named source systems
Materials procurement and inventory	A readiness flag is consumed; the supply chain is not modelled
Financial costing	Costs are expressed in weighted delay minutes, not currency
Cross-division through-traffic coordination	Zonal layer handles machine allocation only

7 Data model overview — the four buckets

Every dataset falls into one of four buckets. Naming the bucket tells you why the data exists and what breaks without it.

Bucket	Question answered	Change rate	Breaks what if missing
Reference	What exists on the ground?	Built once	Merging — cannot tell what is co-located or compatible
Demand	What work is pending?	Daily	Everything — nothing to schedule
Supply	When is the corridor free?	Weekly / daily	Assignment — no windows to schedule into
History	What actually happened last time?	Continuous	All four models — nothing to learn from

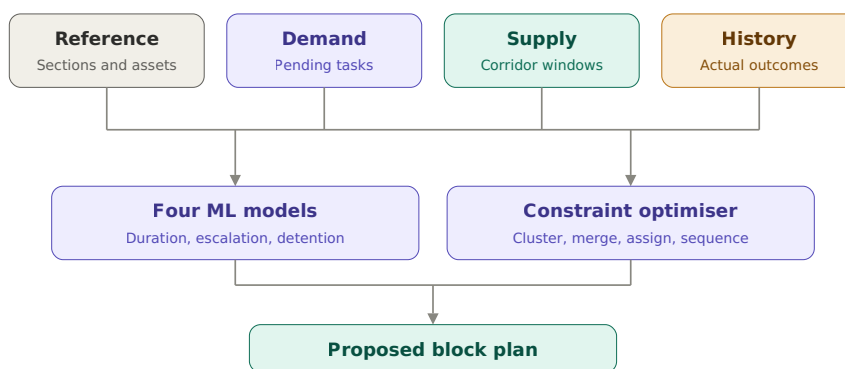


Figure 1 — The four buckets and what consumes them.

Most projects collect demand and supply and stop. **Reference data makes merging possible; history makes the machine learning possible.** Missing those two leaves a calendar application.

7.1 Type conventions

`enum` values are collected in §12.1. `ref` denotes a foreign key. `list[...]` is stored as JSON or a join table. Fields marked ★ are derived by the system and never ingested.

Station code caveat. Codes used as examples (MAS, MSB, MS, TBM, CGL, AJJ, KPD, JTJ, TRL, AVD, GPD, PER, BBQ) follow standard Indian Railways abbreviations for Chennai Division. Verify each against the current Southern Railway working timetable before use in a submitted build — they are illustrative, not authoritative.

8 Bucket 1: Reference data

8.1 Station master

Field	Type	Req	Example / notes
station_code	string	yes	TRL
station_name	string	yes	Tiruvallur
km_from_origin	float	yes	42.0, measured from MAS
is_block_station	bool	yes	Defines block section boundaries
has_yard	bool	yes	Determines whether node-located tasks can occur
route_capacity	int	no	Routes available; a disconnection consumes some
is_junction	bool	yes	true for AJJ, CGL

8.2 Block section master (graph edges)

Field	Type	Req	Example / notes
block_section_id	string	yes	TRL-AJJ-UP
from_station	ref	yes	TRL
to_station	ref	yes	AJJ
line_id	enum	yes	UP — part of the key ; parallel edges share endpoints
start_km / end_km	float	yes	42.0 / 68.5
corridor_id	ref	yes	MAS-AJJ — used only at the monthly horizon
parallel_edges	list[ref]	yes	Lateral adjacency — reroute options
next_sections	list[ref]	yes	Longitudinal adjacency — cascade delay
electrified	bool	yes	false only for the AJJ-Kanchipuram stretch
sectional_speed_kmph	int	yes	110
daily_train_count	int	yes	210
traffic_type	enum	yes	trunk
bidirectional_capable	bool	yes	Can single-line working run here?
monsoon_sensitive	bool	no	true for low-lying suburban stretches
division_id	ref	yes	SR-MAS

8.3 Asset master

Field	Type	Req	Example / notes
asset_id	string	yes	PT-TRL-04
asset_type	enum	yes	point_machine
department	enum	yes	SNT
block_section_id	ref	no	Null for node-located assets
station_code	ref	no	Null for edge-located assets
km	float	yes	42.1
criticality_class	enum	yes	A / B / C
install_date / last_overhaul_date	date	no	Age features
cumulative_tonnage_gmt	float	no	Track assets only; a strong escalation predictor
failure_count_12m	int	no	2

8.4 Task type catalogue

Field	Type	Req	Example / notes
task_type_id	string	yes	ENG-RAIL-WELD
description	string	yes	Rail flaw welding and piece replacement
department	enum	yes	ENG
location_kind	enum	yes	edge / node — determines access mechanism
access_required	enum	yes	traffic_block
nominal_duration_min	int	yes	180
min_crew	int	yes	6
machine_required	enum	yes	none
night_permitted	bool	yes	true
monsoon_restricted	bool	yes	false
adjacent_line_speed_restriction_kmph	int	no	30 — caution imposed on parallel lines during this work
worksite_length_km	float	no	1.0 — length over which the caution applies
default_periodicity_days	int	no	Null for defect-driven work; drives monthly forecast

8.5 Compatibility matrix **HIGHEST VALUE PER ROW**

Thirty to forty hand-authored rules. The only place genuine railway domain knowledge enters the system.

Field	Type	Req	Example / notes
rule_id	string	yes	CMP-018
type_a / type_b	ref	yes	SNT-POINT-SERVICE / ENG-TAMPING
relation	enum	yes	parallel / sequential / incompatible
condition	string	no	Same turnout, or within 200 m
max_distance_m	int	no	200
rule_basis	string	yes	Heavy machine movement over a disconnected point machine
validated_by	string	no	Deliberately blank until an officer confirms it

- **parallel** — may run simultaneously; merged duration takes the maximum
- **sequential** — may share a block but must run in order; durations chain
- **incompatible** — must never share a block

8.6 Machine and crew master

Field	Type	Req	Example / notes
Machines			
machine_id / machine_type	string / enum	yes	TAMP-SR-07 / tamper
home_base / current_section	ref	yes	AVD / MAS-AJJ-02
available_from / available_to	datetime	yes	Availability calendar
output_rate	float	no	Metres per hour; feeds duration estimation
transit_time_h	float	yes	Hard constraint input — hours to reposition per adjacent section
Crews			
crew_id / department	string / enum	yes	ENG-TRL-A / ENG
base_section / size	ref / int	yes	MAS-AJJ-03 / 8
shift_pattern	enum	yes	rotating_3
qualifications	list[ref]	yes	["ENG-RAIL-WELD", "ENG-TAMPING"]
min_rest_hours	int	yes	Hard constraint — minimum between duties
max_consecutive_nights	int	yes	Hard constraint
max_weekly_duty_hours	int	yes	Hard constraint

9 Bucket 2: The unified task schema

The central artefact of the project. All five points of view, all four models and the optimiser read from this one object. Three source systems with three vocabularies collapse into it. Build it first and change it reluctantly.

Field	Type	Req	Notes
Identity			
task_id	string	yes	ENG-2026-04412
source_system	enum	yes	TMS / SMMS / TDMS — kept for traceability and reconciliation
source_ref	string	yes	Original identifier in the source system
department	enum	yes	ENG / TRD / SNT
task_type_id	ref	yes	ENG-RAIL-WELD
asset_id	ref	no	Null for span-based work with no single asset
Location — merging depends entirely on these			
location_kind	enum	yes	edge / node
block_section_id	ref	no	TRL-AJJ-UP — present when location_kind is edge
station_code	ref	no	TRL — present when location_kind is node
start_km / end_km	float	yes	42.10 / 42.15 — equal for point work
line_id	enum	no	UP — null for node work
Origin and urgency			
origin	enum	yes	defect / overdue / routine / emergency_followup / proactive
raised_date	date	yes	2026-09-03
days_pending	int	yes	Recomputed daily
severity	enum	yes	IMDT / 1 / 2 / 3 for rail flaws; critical / major / minor otherwise
deadline	date	no	Hard constraint when safety-critical. Never a soft penalty
safety_critical	bool	yes	Drives the hard deadline constraint
Access and effort			
access_required	enum	yes	traffic_block / power_block / disconnection / none
requested_duration_min	int	yes	180 — what the department asked for
crew_required	int	yes	6
machine_required	enum	yes	none / tamper / BCM / USFD_car / OHE_tower_car
materials_ready	bool	yes	Feeds the availment risk model
Constraints			
season_restricted	bool	yes	true blocks scheduling during the Oct-Dec monsoon
night_permitted	bool	yes	false forces daylight — expensive on suburban lines
status	enum	yes	pending / scheduled / in_progress / done / deferred / cancelled

Derived by the system — never ingested ★			
geo_key	string	★	block_section_id + km bucket, for fast clustering
predicted_duration_p50 / p80 / p95	int	★	Quantile outputs; buffer = p80 – p50
escalation_hazard_daily	float	★	Calibrated marginal daily hazard
availment_risk	float	★	Probability the block is wasted
risk_score	float	★	Composite ranking shown to officers
score_factors	list	★	Named factor contributions — required for explainability

9.1 Sample records — the scenario from Document 3

```
{ "task_id": "ENG-2026-04412", "source_system": "TMS", "department": "ENG",
  "task_type_id": "ENG-RAIL-WELD", "asset_id": "RL-TRLAJJ-0421",
  "location_kind": "edge", "block_section_id": "TRL-AJJ-UP",
  "start_km": 42.10, "end_km": 42.15, "line_id": "UP",
  "origin": "defect", "raised_date": "2026-09-03", "days_pending": 4,
  "severity": "1", "safety_critical": true, "deadline": "2026-09-17",
  "access_required": "traffic_block", "requested_duration_min": 180,
  "crew_required": 6, "machine_required": "none", "materials_ready": true,
  "season_restricted": false, "night_permitted": true, "status": "pending" }
```

```
{ "task_id": "TRD-2026-01188", "source_system": "TDMS", "department": "TRD",
  "task_type_id": "TRD-OHE-INSP", "asset_id": "OHE-TRLAJJ-0402",
  "location_kind": "edge", "block_section_id": "TRL-AJJ-UP",
  "start_km": 40.00, "end_km": 44.00, "line_id": "UP",
  "origin": "routine", "raised_date": "2026-08-28", "days_pending": 10,
  "severity": "minor", "safety_critical": false, "deadline": null,
  "access_required": "power_block", "requested_duration_min": 90,
  "crew_required": 4, "machine_required": "OHE_tower_car", "materials_ready": true,
  "season_restricted": false, "night_permitted": true, "status": "pending" }
```

```
{ "task_id": "SNT-2026-00734", "source_system": "SMMS", "department": "SNT",
  "task_type_id": "SNT-POINT-SERVICE", "asset_id": "PT-TRL-04",
  "location_kind": "node", "station_code": "TRL",
  "start_km": 42.10, "end_km": 42.10, "line_id": null,
  "origin": "overdue", "raised_date": "2026-08-27", "days_pending": 11,
  "severity": "major", "safety_critical": false, "deadline": "2026-09-30",
  "access_required": "disconnection", "requested_duration_min": 60,
  "crew_required": 3, "machine_required": "none", "materials_ready": true,
  "season_restricted": false, "night_permitted": true, "status": "pending" }
```

These three records are the merge candidate the optimiser must discover. Use them as the first integration test.

10 Bucket 3: Supply data

10.1 Train paths

Field	Type	Req	Example / notes
<code>path_id</code>	string	yes	P-12639-TRLAJJ-UP
<code>train_no</code> / <code>train_type</code>	string / enum	yes	12639 / superfast
<code>block_section_id</code>	ref	yes	TRL-AJJ-UP
<code>entry_time</code> / <code>exit_time</code>	time	yes	06:12 / 06:24
<code>days_of_week</code>	string	yes	"1111100"
<code>priority_class</code>	int	yes	1 (highest) to 6 (goods) — weights detention cost
<code>divertible</code>	bool	yes	Can it use a parallel line during a block?

10.2 Goods traffic forecast

Field	Type	Req	Example / notes
<code>forecast_date</code>	date	yes	2026-09-10
<code>block_section_id</code>	ref	yes	BBQ-GPD-UP — Ennore and Chennai Port traffic
<code>time_band</code>	enum	yes	00-06 / 06-12 / 12-18 / 18-24
<code>expected_rakes</code>	int	yes	7
<code>confidence</code>	float	no	Converts an apparent gap into a probabilistic one

10.3 Corridor windows

Generated from the documented pattern in Blueprint §5.4; replaced on deployment by the division's actual corridor block table.

Field	Type	Req	Example / notes
<code>window_id</code>	string	yes	W-TRLAJJ-UP-WED
<code>block_section_id</code>	ref	yes	TRL-AJJ-UP
<code>day_of_week</code>	enum	yes	WED
<code>start_time</code> / <code>duration_min</code>	time / int	yes	01:30 / 180
<code>window_type</code>	enum	yes	corridor_block / traffic_gap / requested
<code>max_departments</code>	int	no	Practical ceiling on concurrent departments

11 Bucket 4: Historical data

The bucket nobody currently has, and the reason the field application exists. No system in service records how long a job actually took against how long it was requested for. Collecting it is the project's most defensible contribution, independent of the optimiser.

11.1 Block execution history → duration and availment models

Field	Type	Req	Notes
block_id	string	yes	BLK-2026-00891
block_section_id	ref	yes	Where
tasks_included	list[ref]	yes	One for a single job, several for a merged block
departments	list[enum]	yes	Measures coordination directly
merge_group_id	string	no	Cross-reference used in the BDMS handoff
requested_start / requested_duration_min	datetime / int	yes	What was asked for
sanctioned_start / sanctioned_duration_min	datetime / int	yes	What was granted
actual_start / actual_end	datetime	yes	The target variable
availed	bool	yes	Label for the availment model
overrun_min	int	yes	Actual minus sanctioned; may be negative
overrun_reason	enum	no	material_delay / crew_short / machine_late / weather / unexpected_condition
crew_id / machine_id	ref	no	Model features
weather / season	enum	yes	clear / rain / heavy_rain; normal / monsoon
was_merged	bool	yes	Essential feature — merged work is often slower per task

11.2 Defect lifecycle → escalation risk model

Expanded at training time into one row per defect-day for discrete-time survival modelling.

Field	Type	Req	Notes
defect_id / asset_id	string / ref	yes	Joins to asset master for tonnage and age
defect_type / severity_at_raise	enum	yes	
raised_date / attended_date	date	yes	attended_date null if still open. Never a model feature — leakage
days_open	int	yes	Hazard typically rises with age
temporary_repair_applied	bool	no	Materially changes hazard
escalated	bool	yes	The label
escalation_type / escalation_date	enum / date	no	speed_restriction / failure / punctuality_incident

11.3 Detention log → detention residual model

Field	Type	Req	Notes
block_id	ref	yes	Joins to execution history
analytical_estimate_min	int	yes	Feature input to the residual model
trains_affected	int	yes	
total_detention_min	int	yes	Target — and the unit of the objective function
rerouted_count / cancelled_count	int	no	
time_band	enum	yes	Peak versus off-peak changes everything

11.4 Emergency event log

Field	Type	Req	Notes
event_id / event_type	string / enum	yes	rail_fracture / OHE_failure / point_failure / cattle_run_over / weather
block_section_id / km	ref / float	yes	
occurred_at / duration_min	datetime / int	yes	How long the section was seized
resources_consumed	list[ref]	yes	Drives the re-plan — usually the larger disruption
blocks_invalidated	list[ref]	no	Planned work that lost its window or crew
followup_task_id	ref	no	Permanent repair generated with a short hard deadline

12 Enumerations, mappings, volumes and file layout

12.1 Enumeration reference

Enumeration	Values
department	ENG, TRD, SNT
source_system	TMS, SMMS, TDMS
location_kind	edge, node
asset_type	rail, sleeper, ballast, turnout, bridge, level_crossing, OHE_span, mast, insulator, contact_wire, substation, signal, point_machine, track_circuit, axle_counter, interlocking
access_required	traffic_block, power_block, disconnection, none
origin	defect, overdue, routine, emergency_followup, proactive
severity (rail flaws)	IMDT, 1, 2, 3
severity (general)	critical, major, minor
criticality_class	A, B, C
line_id	UP, DN, UP_SUB, DN_SUB, THIRD, FOURTH, SINGLE
traffic_type	suburban, trunk, freight, mixed, branch
machine_required	none, tamper, BCM, USFD_car, OHE_tower_car, ballast_regulator, rail_grinder
train_type	rajdhani_class, superfast, express, passenger, EMU, MEMU, goods, special, light_engine
relation	parallel, sequential, incompatible
status	pending, scheduled, in_progress, done, deferred, cancelled
block_state	DRAFT, DEPT_REVIEW, OPS_REVIEW, SANCTIONED, ISSUED, IN_PROGRESS, COMPLETED, ABANDONED, REJECTED, REFUSED
rejection_reason	crew_unavailable, material_not_received, duration_insufficient, wrong_sequence, machine_unavailable, traffic_conflict, other
overrun_reason	material_delay, crew_short, machine_late, weather, unexpected_condition, traffic_hold, none
escalation_type	speed_restriction, failure, punctuality_incident
event_type	rail_fracture, OHE_failure, point_failure, cattle_run_over, weather, signal_failure, other
time_band	00-06, 06-12, 12-18, 18-24
season	normal, monsoon
window_type	corridor_block, traffic_gap, requested

12.2 Dataset to component mapping

Use this to sequence the build: if a component's input tables are not ready, that component cannot start.

Component	Reads from	Produces
Duration predictor	11.1, task type catalogue, machine and crew master	P50 / P80 / P95 minutes
Escalation risk	11.2 expanded to defect-days, asset master	Calibrated daily hazard
Detention estimator	11.3, train paths, block section master (line count decisive)	Weighted detention minutes
Availment risk	11.1, machine master, task materials_ready	Probability the block is wasted
Window enumerator	Train paths, goods forecast, corridor windows	Candidate windows, flagged by type
Clustering	Task pool, compatibility matrix, block section master	Merge candidates with critical-path duration
Weekly optimiser	All of the above, plus crew and machine constraints	Weekly block plan
Monthly optimiser	Periodicity, defect arrival rates, machine master	Quotas, machine positions, reserved capacity
Emergency re-planner	11.4, current plan, resource state	Revised plan plus diff

12.3 Real, synthetic and expert-validated

Category	Approach
Ground in reality	Section and station names and codes, kilometre markers, line counts, electrification status, approximate train frequencies and suburban headways, the Oct-Dec northeast monsoon window, port-bound freight patterns on the northern line
Synthesise	Individual tasks, defect arrival rates by asset type, historical block executions, actual-versus-planned durations with a deliberate positive bias , emergency events at plausible frequency, crew and machine calendars
Expert validation required	The compatibility matrix, and the corridor block pattern. Both are authored approximations, both flagged, both replaced on deployment

Cold-start honesty. On day one the duration model has no history. The correct behaviour is to fall back to requested duration plus a fixed buffer and say so. Concealing this is worse than admitting it — an evaluator who spots it unprompted will discount everything else.

12.4 Volume estimates

Dataset	Rows (pilot, 1 year)	Notes
Station master	~30 (pilot) / 160 (division)	Static
Block section master	~60 edges (pilot)	Multigraph — multiple edges per station pair
Asset master	2,000 – 5,000	Static
Task type catalogue	30 – 50	Authored
Compatibility matrix	30 – 40	Authored — highest value per row
Machine / crew master	~50 / ~120	Static
Maintenance tasks	~4,000	~300 pending in any week
Train paths	~15,000	Largest table by row count
Corridor windows	~200	Weekly pattern
Block execution history	~2,000	The training set
Defect lifecycle	~3,000	Expands to ~25,000 defect-day rows
Detention log	~2,000	One per executed block
Emergency log	~150	

Total under 50 MB. This fits one PostgreSQL instance and loads entirely into a pandas dataframe. There is no case for distributed streaming infrastructure at this scale, and introducing it signals a misreading of the problem. Stating the volume confidently is itself a credibility signal.

12.5 Repository file layout

```
data/
  reference/
    stations.csv           8.1
    block_sections.csv     8.2  <-- graph edges, keyed by line
    assets.csv             8.3
    task_types.csv         8.4
    compatibility_matrix.csv 8.5  <-- author by hand, carefully
    machines.csv  crews.csv 8.6
  demand/
    tasks.csv             Section 9 - the unified schema
  supply/
    train_paths.csv        10.1
    goods_forecast.csv     10.2
    corridor_windows.csv   10.3  <-- generated from the documented pattern
  history/
    block_executions.csv   11.1
    defect_lifecycle.csv   11.2
    detention_log.csv       11.3
    emergency_events.csv   11.4
  generator/
    chennai_geography.py   real section and station constants
    corridor_pattern.py    the documented corridor block rule set
    generate_all.py        emits every CSV above with a fixed seed
```

Use a fixed random seed so every team member and every demonstration run works against identical data. Nothing undermines a live demonstration faster than numbers that shift between rehearsal and presentation.

Build order. Reference data first, because clustering depends on it. Then the unified task schema, because everything reads from it. Then supply data, so the optimiser has windows to assign into. History last — it is needed only once model training begins, and it is easiest to generate once the other three buckets define the shape of a valid record.